

MinIO: Cloud Storage Solution Overview

1. What is MinIO?

MinIO is an **open-source object storage server** that is **compatible with Amazon S3 APIs**. It is designed to store unstructured data like images, videos, backups, and logs. Unlike public cloud services (AWS S3, Google Cloud Storage), **MinIO is software you install and manage**, which means you control where your data lives.

Key Features:

- High-performance object storage for large-scale data.
- Scalable from a single server to distributed multi-node clusters.
- Supports **erasure coding, replication, and high availability**.
- Cloud-native API compatibility (S3-compatible), so apps built for S3 can work with MinIO.

2. Will images be stored in the cloud?

It depends entirely on **where you host MinIO**:

Hosting Location	Storage Location	Notes
Local PC / On-premises server	On your local machine	Data stays local; not publicly accessible unless networked.
Cloud server (AWS, GCP, Azure)	On the cloud provider's VM disk	Data is effectively in the cloud; accessible over the internet.
Hybrid / Distributed	Across multiple servers/disks	Can replicate to cloud and on-prem; very scalable and reliable.

Conclusion:

- **MinIO itself does not automatically store data in the cloud.**
- If you want “cloud storage,” you must **host MinIO on a cloud server** or **sync it to cloud storage**.

3. How much data can be stored?

MinIO is **highly scalable**, but storage limits depend on your hardware setup:

1. **Single server setup:**
 - Limited by the server's total disk capacity.
 - Example: A server with 4 TB of disk can store ~4 TB of images.
2. **Distributed / multi-node setup:**
 - You can combine multiple servers and disks to store **dozens of TBs or even PBs**.
 - Supports **erasure coding**, which protects data while spreading it across nodes.

Practical Example for Your Use Case:

- Storing **100–1000 GB** of comic images is **easily achievable** on a single moderately-sized server.
- For **massive storage or enterprise-scale projects**, a distributed MinIO cluster is recommended.

4. Summary

- **MinIO** = S3-compatible object storage software.
- **Cloud storage?** Only if hosted on a cloud server; otherwise, data stays local.
- **Data volume:** From hundreds of GBs to petabytes, depending on server/disk setup.
- **Use case:** Perfect for projects like storing comic images, backups, media libraries, or private cloud data storage.